

Assignment: CNN Development on Custom Dataset

1. Assignment Overview

- Objective: Design and train a CNN to classify images from selected dataset
- Dataset: Select open access dataset from **Kaggle** or **Hugging Face** etc. with multiple classes
- Submission Format: **CNN_StudentID.ipynb** (GitHub repository link and a copy of the file)
- **Total Marks: 10**
- **Due Date: July 27, 2026 (Monday)**

2. Assignment Requirements

Data Loading and Preprocessing

- Load dataset
- Create training and validation data loaders with batch size

Model Architecture

- Build a custom CNN
- Show Document architecture with a detailed summary

Training and Optimization

- Use appropriate optimizer (Adam, SGD)
- Use cross-entropy loss for multi-class classification
- Plot training and validation loss/accuracy curves

Model Evaluation

- Evaluate on test set and report comprehensive metrics
- Generate confusion matrix visualization
- Analyze per-class performance (best and worst performing classes)
- Save trained model weights

Code Quality & Documentation

- Well-organized, readable code with proper comments
- python notebook with explanations for each section
- Include hyperparameter details and rationale

3. Evaluation Criteria (10 Marks)

Criteria	Description	Marks
Model Architecture	Appropriate CNN design with clear justification	2
Data Handling	Correct loading, preprocessing	1
Training	Successful training with optimization & regularization	2
Test Accuracy	Classification performance on test set	2
Evaluation Metrics	Comprehensive metric reporting (Precision, Recall, F1)	1
Visualizations	Accuracy and Loss curves, confusion matrix	1
Code Quality, Documentation and Saved Model	Well-written, commented code, saved model	1

4. Suggested Notebook Structure

1. Import Libraries
2. Load and Explore Dataset
3. Data Preprocessing & Augmentation
4. Define CNN Architecture
5. Training Loop with Validation
6. Evaluate Model on Test Set
7. Visualizations (Loss/Accuracy Curves, Confusion Matrix)
8. Analysis & Discussion of Results
9. Conclusions & Future Work

5. GitHub Should Contain these Deliverables

1. **Python Notebook (.ipynb)**: Complete implementation with code, outputs, and visualizations.
Name: CNN_StudentID.ipynb
2. **Model Weights**: Save trained model in .pth [If the model is more than 20MB do not upload]

6. A Microsoft Form will be shared later for the submission of GitHub link and a copy of your ipynb.

7. Resources:

Model Save and Load Example

https://github.com/Rifath-Mahmud/CVPR/blob/main/Model_Save_and_load_Example.ipynb

CNN Model Example:

https://github.com/Rifath-Mahmud/CVPR/blob/main/catdog_cnn.ipynb